

Sr⁺ ionisation limit 45932.20 cm⁻¹ = 5.69487 eV (217.712 nm)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$5s^2\ ^3P_0 - 5s6d\ ^3D_1$	394.1924	394.0808	760.5232	-	-	-	-	394 nm
$5s^2\ ^3P_0 - 5s6d\ ^3D_3$	403.1516	403.0377	743.6221	-	-	-	-	403.15 nm
$5s^2\ ^3P_0 - 5s6d\ ^3D_2$	403.3517	403.2378	743.2532	-	-	-	-	403.35 nm repumper
$5s^2\ ^3P_0 - 5s6d\ ^3D_1$	403.4326	403.3187	743.1041	-	-	-	-	403.43 nm
$5s^2\ ^1S_0 - 5s5p\ ^1P_1$	460.8624	460.7333	650.5032	5.252	1.903e+08	30.51	100	461 nm blue MOT / Zeeman slower / imaging
$5s^2\ ^3P_0 - 5p^2\ ^3D_1$	474.3248	474.1922	632.0404	2.482	3.900e+07	-	-	474 nm
$5s^2\ ^3P_0 - 5s5d\ ^1D_1$	483.3393	483.2042	620.2526	2.349	3.300e+07	-	-	483 nm repumper
$5s^2\ ^3P_0 - 5s5d\ ^3D_3$	496.3648	496.2263	603.9761	5.093	6.140e+07	-	-	496.36 nm
$5s^2\ ^3P_0 - 5s5d\ ^3D_1$	496.0328	496.7942	603.2857	1.969	1.280e+07	-	-	496.93 nm repumper
$5s^2\ ^3P_0 - 5s5d\ ^1D_1$	497.3055	497.1668	602.8336	0.487	1.300e+06	-	-	497.30 nm
$5s^2\ ^1S_0 - 5s5p\ ^3P_0$	671.2065	671.0122	446.6471	-	1.520e+04	3.183e-10	7.6	671 nm magnetic-quadrupole line
$5s^2\ ^3P_0 - 5s6s\ ^3S_1$	679.2894	679.1020	441.3324	1.970	8.360e+06	11.43	11.6	679 nm repumper
$5s^2\ ^3P_1 - 5s6s\ ^3S_1$	688.0211	687.8313	435.7315	3.437	2.450e+07	11.43	34.1	688 nm
$5s^2\ ^1S_0 - 5s5p\ ^3P_1$	689.4491	689.2589	434.8290	0.151	4.700e+04	0.007479	10	689 nm red MOT
$5s^2\ ^1S_0 - 5s5p\ ^3P_0$	698.4457	698.2531	429.2280	-	1.349e-09	-	-	698 nm optical clock / optical qubit
$5s^2\ ^3P_0 - 5s6s\ ^3S_1$	707.1995	707.0046	423.91497(3)	4.525	3.910e+07	11.43	54.3	707 nm repumper
$5s^2\ ^3P_1 - 5s4d\ ^1D_2$	1771.3690	1770.8854	169.2434	0.164	1.950e+03	0.0003772	82	1D2 decay returning atoms to the ground state via 3P1
$5s^2\ ^3P_0 - 5s4d\ ^1D_2$	1904.3484	1903.8286	157.4252	0.084	1.800e+02	0.0003772	17.7	1D2 decay to the metastable reservoir state

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (mHz)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	394.192	88-04	91(0.54)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	394.192	88-86	54(60.3)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	394.192	88-87	51(64.28)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.152	88-84	85(10)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.152	88-86	41(10)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.152	88-87	53(5)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.352	88-84	81(0.20)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.352	88-86	49(1.02)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.352	88-87	20(5)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.433	88-84	77(10)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.433	88-86	30(10)
$5s5p\ ^3P_2 - 5s6d\ ^3D_1$	403.433	88-87	18(0.20)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1$	460.862	84-88	-275(3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1$	460.862	86-88	-126(0.20)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1$	460.862	87-88	-47(0.20)
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	483.339	88-84	90.00
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	483.339	88-86	40.00
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	483.339	88-87	32.00
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	496.365	88-84	91(60.20)
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	496.365	88-86	46(20.0)
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	496.365	88-87	27(0.20)
$5s5p\ ^3P_2 - 5s5d\ ^3D_1$	496.933	88-84	92(80.20)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || er || i \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)

